ECA13 – MICROWAVE ENGINEERING

CO1 APPLICATION-BASED ASSESSMENT

Analysis, Design and Verification of an X-Band Radar Waveguide Feed Network

Student Name: _____ Register No.: _____
Note on engineering values: This report is a worked design example. Analytical values are calculated from stated assumptions; loss and S-parameter figures identified as estimates/ targets must be verified using a field solver, VNA measurement, or manufacturer data before hardware acceptance.

Abstract

This report designs an X-band transmission path for a civil weather-radar front end. A standard WR-90 rectangular waveguide is selected to carry a 10 GHz signal between the transmitter/receiver module and antenna feed. Its dominant TE10 mode, cutoff limits, guide wavelength, phase/group velocities, impedance, loss budget and two-port network behaviour are evaluated. A practical acceptance specification, matching approach, tolerance analysis and EMC test plan are provided. Calculations are first-order design estimates and require simulation or measurement for final verification.

Stage 1 – Application and Quantified Requirements (10 marks)

1.1 Application and microwave history

Application: civil weather-radar front-end feed network operating at 10 GHz in the X band. The waveguide carries microwave power between the RF front end and antenna feed with low conductor loss, high power capability and good shielding. Maxwell's electromagnetic theory established the field framework; Hertz demonstrated electromagnetic waves; cavity magnetrons accelerated high-power radar development; later satellite links, microwave integrated circuits (MMICs), and modern 5G/6G systems expanded microwave use. Weather radar benefits from directional antennas and X-band resolution, although rain attenuation is higher than at lower frequencies.

1.2 Frequency-band comparison

Band	Example frequency	Free-space wavelength	Advantages	Limitations
S band	3 GHz	100 mm	Lower rain attenuation; useful range	Larger components/ apertures for a given electrical size
X band	10 GHz	30 mm	Compact feed and antenna; useful bandwidth and radar resolution	More rain attenuation than S band
Ku band	15 GHz	20 mm	Compact hardware and broad available spectrum in many	Greater atmospheric/ rain attenuation; more demanding tolerances

Selected operating frequency: 10.0 GHz; design operating band: 9.5–10.5 GHz (1.0 GHz bandwidth). The WR-90 guide itself is commonly specified for 8.2–12.4 GHz, so the selected band lies within its single-mode operating region.

1.3 Quantified requirement specification

Parameter	Requirement / acceptance target
Application category	Civil weather radar
Operating frequency/band	10.0 GHz nominal; 9.5–10.5 GHz
Structure/mode	WR-90 rectangular waveguide; TE10 dominant mode
Characteristic/equivalent impedance	TE10 wave impedance about 506 Ω at 10 GHz (not a 50 Ω coaxial impedance)
Maximum insertion loss	≤0.20 dB for 0.30 m guide section including transitions
Input return loss	≥20 dB; VSWR ≤1.22
Port isolation / shielding	≥60 dB shielding effectiveness target for enclosure; two-port isolation not applicable to a simple through-feed
Power handling	Initial design target: 10 W average; peak-power rating to be confirmed from flange, arcing and breakdown analysis
Size	WR-90 internal a=22.86 mm, b=10.16 mm; guide length 0.30 m
Cost target	Prototype feed section and transitions ≤₹8,000 (indicative, supplier dependent)
Environmental/safety	Grounded enclosure, weather sealing, no exposed RF joints; comply with applicable RF exposure rules

Constraints: avoid leakage at flanges, maintain electrical contact and alignment, use suitable waveguide-rated components, and evaluate personnel exposure at the antenna. The 10 W target applies to the feed section, not the radiated exposure limit; antenna power density/SAR assessment is application-specific.

Stage 2 – Evaluate and Select the Transmission Structure (15 marks)

Criterion	Weight	Coaxial	Rectangular waveguide	Stripline	Microstrip
Frequency / mode suitability	20%	4	5	4	4
Loss at X band	20%	3	5	4	3
Power handling	15%	3	5	4	2
Size / weight	10%	4	2	3	5
Cost	10%	4	2	3	5
Shielding / EMI	15%	4	5	5	2
Integration / assembly	10%	4	3	4	5

Weighted score / 5	100%	3.65	4.15	3.95	3.55

Scores are engineering judgements on a 1–5 scale, with 5 best; weighted score = Σ(weight × score)/100. The waveguide wins because low loss, power handling and shielding are more important than size and cost for this radar feed.
Modes: coaxial line supports TEM with no cutoff, but higher-order modes can arise above its design range. Stripline supports TEM because its fields are contained between two ground planes in a homogeneous dielectric. Microstrip is quasi-TEM because fields occupy both substrate and air, so the effective permittivity is frequency-dependent and a pure TEM solution is not exact. A hollow single-conductor waveguide cannot support TEM: a TEM mode requires both longitudinal E and H components to be zero and a two-conductor return path; a hollow guide instead supports TE/TM modes with cutoff.
Sensitivity: if compactness and cost receive much larger weights, microstrip or coax may rank first. If frequency is reduced substantially and power handling becomes less important, coaxial cable may be preferred. At higher frequencies or higher power, waveguide's loss and shielding advantages become more decisive. Lowering frequency toward 5 GHz puts operation below the WR-90 TE10 cutoff; doubling to 20 GHz allows more higher-order modes, so the present single-mode design no longer applies.
Design Decision 1: I select WR-90 rectangular waveguide over microstrip because the 10 GHz feed prioritizes low loss, power handling and EMI shielding. Its TE10 mode propagates between 8.2 and 12.4 GHz, covering the chosen band.

Stage 3 – Mode and Propagation Model (20 marks)

3.1 Transmission-line equations

For a uniform line, $\partial V/\partial z = -(R + j\omega L)I$ and $\partial I/\partial z = -(G + j\omega C)V$. Therefore $\gamma = \alpha + j\beta = \sqrt{[(R+j\omega L)(G+j\omega C)]}$ and $Z0 = \sqrt{[(R+j\omega L)/(G+j\omega C)]}$. For a low-loss line (R ωL and G ωC), $\alpha \approx R/2\sqrt{(L/C)} + G\sqrt{(L/C)}/2$ and $\beta \approx \omega\sqrt{(LC)}$. The voltage reflection coefficient at a load is $\Gamma = (ZL-Z0)/(ZL+Z0)$. These distributed-line equations provide the network framework; a waveguide is described using its modal propagation constant and equivalent TE impedance.

3.2 Waveguide dimensions and cutoff

Choose standard WR-90 dimensions: broad wall a = 22.86 mm, narrow wall b = 10.16 mm; air-filled guide, μr≈1, εr≈1. For a rectangular waveguide, $fc(mn) = (c/2)\sqrt{[(m/a)^2+(n/b)^2]}$. For TE10, $fc10 = c/(2a) = 3.00\times10^8/(2\times0.02286) = 6.56$ GHz. The next cutoffs are TE20 = c/a = 13.11 GHz; TE01 = c/(2b) = 14.76 GHz; TE11/TM11 = $(c/2)\sqrt{[(1/a)^2+(1/b)^2]}$ ≈16.15 GHz.
Thus the first higher-order cutoff is TE20 at 13.11 GHz. A conservative single-mode range is 6.56–13.11 GHz; the manufacturer's WR-90 recommended band is 8.2–12.4 GHz. The chosen 9.5–10.5 GHz band remains below the next-mode cutoff with at least 2.61 GHz margin at the upper band edge.

3.3 Worked propagation calculation at 10 GHz

Quantity	Formula	Calculated value at 10 GHz
Free-space wavelength	$\lambda0 = c/f$	29.98 mm
TE10 cutoff	$fc10 = c/(2a)$	6.56 GHz
Guide wavelength	$\lambda g = \lambda0 / \sqrt{[1-(fc/f)^2]}$	39.71 mm

Phase constant	$\beta = 2\pi/\lambda g$	158.24 rad/m
Phase velocity	$v_p = c / \sqrt{[1-(f_c/f)^2]}$	3.971×10^8 m/s
Group velocity	$v_g = c\sqrt{[1-(f_c/f)^2]}$	2.263×10^8 m/s
TE wave impedance	$Z_{TE} = \eta_0 / \sqrt{[1-(f_c/f)^2]}$	499.0 Ω
Group delay over 0.30 m	$\tau_g = L/v_g$	1.33 ns

The phase velocity exceeds c in the guide, but this does not imply superluminal information transfer: group velocity is below c and carries the signal envelope. TE modes have Ez=0 and generally nonzero Hz; TM modes have Hz=0 and generally nonzero Ez. TEM has Ez=Hz=0 and no cutoff, but is not supported by a hollow rectangular guide.
Higher-order mode risk: discontinuities, poor flange alignment, oversized transitions and operation near the TE20 cutoff can excite unwanted modes. Keep the operating band below 12.4 GHz, use a properly designed taper/launcher, maintain alignment, and avoid abrupt steps. A ±5% error in a changes fc10 by approximately 5%; the nominal cutoff could shift by about 0.33 GHz, still well below 9.5 GHz. A full dimensional tolerance study should also include b and transition geometry.
Design Decision 2: the dominant mode is TE10 with cutoff 6.56 GHz; the practical single-mode band is 8.2–12.4 GHz. WR-90 dimensions are acceptable for the 9.5–10.5 GHz operating band.

Stage 4 – Loss Budget and Loss Reduction (15 marks)

For copper at 10 GHz, use conductivity $\sigma \approx 5.8 \times 10^7$ S/m. Skin depth $\delta = \sqrt{[2/(\omega\mu_0\sigma)]} \approx 0.66$ µm and surface resistance $R_s = \sqrt{[\pi f \mu_0/\sigma]} \approx 0.026$ Ω/square. Actual waveguide attenuation depends on wall finish, plating, joints and mode field distribution; the values below are a preliminary budget for a 0.30 m guide section, not certified manufacturer data.

Loss mechanism	**Initial estimate (dB/m)**	**Improved estimate (dB/m)**	**Reduction method / trade-off**
Conductor	0.12	0.07	Smooth copper/silver-plated interior; higher cost and plating control
Dielectric	0.00	0.00	Air-filled guide; no dielectric liner
Radiation / leakage	0.02	0.01	Improve flange contact and alignment; tighter assembly
Connector / transition equivalent	0.20 per section	0.08 per section	Tapered, tuned transition; requires design/ measurement
Guide-only total attenuation	0.14	0.08	0.30 m guide contribution = 0.042 dB initially, 0.024 dB improved

For a clear total insertion-loss estimate, the 0.30 m guide contributes approximately 0.14×0.30 = 0.042 dB initially. Adding estimated transitions of 0.20 dB gives 0.242 dB, above the 0.20 dB requirement. The improved guide contribution is 0.08×0.30 = 0.024 dB; with improved transitions of 0.08 dB, total is 0.104 dB, which meets the target with 0.096 dB margin. These transition figures are assumed design targets and must be verified.
Initial component percentages of the 0.242 dB total: conductor 0.036 dB (14.9%); radiation/

leakage 0.006 dB (2.5%); transitions 0.200 dB (82.6%). Therefore transition/mismatch-related loss dominates the initial budget. Improved total: 0.021 dB conductor, 0.003 dB leakage and 0.080 dB transitions = 0.104 dB approximately.
Conductor-loss trend: Rs rises approximately as √f, so conductor loss generally increases with frequency. Higher conductivity lowers Rs; rough surfaces increase effective resistance. Dielectric loss rises with frequency and loss tangent, but an air-filled guide largely avoids substrate dielectric loss. Use smooth, conductive interior plating and a low-reflection tapered transition. The trade-off is increased manufacturing cost and tighter assembly tolerances.
Design Decision 3: the dominant initial loss is transition loss, approximately 82.6% of the total. Improved transitions and flange contact reduce estimated total insertion loss from 0.242 dB to 0.104 dB.

Stage 5 – Network Model, Matching and S-Parameters (20 marks)

5.1 Equivalent voltage/current and wave variables

For non-TEM waveguides, voltage and current are not uniquely defined by a cross-sectional electrostatic potential and a single conductor current. An equivalent modal voltage/current can nevertheless be defined by power normalization. For TE modes the modal wave impedance is $Z_{TE} = \omega\mu/\beta = \eta_0/\sqrt{[1-(f_c/f)^2]}$. At 10 GHz, $Z_{TE} \approx 506\ \Omega$. The chosen reference impedance for the modal network is therefore 506 Ω at 10 GHz; a coaxial 50 Ω instrument requires a designed transition and calibration plane.
For a real reference impedance Z_0, normalized waves are $a=(V+Z_0I)/(2\sqrt{Z_0})$ and $b=(V-Z_0I)/(2\sqrt{Z_0})$. The scattering relation is $[b_1\ b_2]^T = [S_{11}\ S_{12};\ S_{21}\ S_{22}][a_1\ a_2]^T$. S_{11} and S_{22} describe input/output reflection; S_{21} and S_{12} describe forward/reverse transmission. Return loss = $-20\log_{10}|S_{11}|$ and insertion loss = $-20\log_{10}|S_{21}|$.

5.2 Two-port representation

For a uniform line section with characteristic impedance Z_c, propagation constant γ and length l, the ABCD matrix is $[[\cosh(\gamma l),\ Z_c \sinh(\gamma l)],\ [(1/Z_c)\sinh(\gamma l),\ \cosh(\gamma l)]]$. Two sections are cascaded by multiplying their ABCD matrices in physical order. For a reciprocal passive guide section, $AD-BC=1$ and $S_{21}=S_{12}$ when both ports use consistent reference impedances. For a lossless section, $|S_{11}|^2+|S_{21}|^2=1$; with loss, the sum is less than one. A symmetric section has $S_{11}=S_{22}$.
For a simple load illustration, take a modal load $Z_L=600\ \Omega$ at the 10 GHz reference plane and $Z_0=506\ \Omega$. $\Gamma=(600-506)/(600+506)=0.0850$. $VSWR=(1+|\Gamma|)/(1-|\Gamma|)=1.186$. Return loss$=-20\log_{10}(0.0850)=21.4$ dB, which meets the ≥20 dB target but has limited margin.

5.3 Matching arrangement and target S-parameters

Use a tapered waveguide-to-load transition or a tuned matching section designed in a full-wave solver. For a simplified equivalent real-impedance example, a quarter-wave transformer has $Z_t=\sqrt{(Z_0 Z_L)}=\sqrt{(506\times600)}=551\ \Omega$ and electrical length $\lambda_g/4 \approx 7.50$ mm at 10 GHz. This is a modal equivalent illustration; the physical waveguide transition must be optimized in EM simulation rather than fabricated as a literal 551 Ω waveguide section.

Parameter	Before matching (illustrative)	After matching target	Interpretation
Reference impedance	506 Ω modal	506 Ω modal	At 10 GHz
|Γ| / VSWR	0.085 / 1.186	≤0.056 / ≤1.12	Target return loss ≥25 dB

Return loss	21.4 dB	≥25 dB	Lower reflection after matching
Insertion loss, 0.30 m plus transitions	0.242 dB estimate	0.104 dB estimate	Target ≤0.20 dB
S11 magnitude	0.085	≤0.056	|S11|=10^(−RL/20)
S21 magnitude	≈0.972 for 0.242 dB power insertion loss	≈0.988 for 0.104 dB	Magnitude estimated from insertion loss
Reciprocity	Expected for passive reciprocal guide	S21≈S12	Verify with VNA
Lossless check	Not lossless	Not lossless	Conductor/transition losses make |S11|²+|S21|²<1

Verification: sweep a calibrated vector network analyser from 9.5 to 10.5 GHz with waveguide calibration standards or suitable coax-to-waveguide calibration. Record S11, S21, S12 and S22 magnitude/phase. Independently calculate cutoff and propagation values using the equations in Stage 3; compare with an EM solver such as CST, HFSS or an equivalent tool. The S-parameter values above are acceptance targets/estimates, not measured plots.

Stage 6 – Validation, Critique and Tolerance (10 marks)

Requirement	**Specified**	**Initial design estimate**	**Final design estimate / verdict**
Operating band / mode	9.5–10.5 GHz; single mode	Within WR-90 recommended band	Pass
Dominant mode / cutoff	TE10; fc=6.56 GHz	TE10; next cutoff 13.11 GHz	Pass with 2.61 GHz margin above 10.5 GHz
Insertion loss	≤0.20 dB	0.242 dB	0.104 dB; Pass (estimated)
Impedance / VSWR	VSWR ≤1.22	1.186 example load	≤1.12 target; Pass if verified
Return loss	≥20 dB	21.4 dB example load	≥25 dB target; Pass if verified
Shielding / isolation	≥60 dB enclosure target	Not measured	Test enclosure leakage; pending
Power / size	10 W average; WR-90, 0.30 m	Geometry meets size	Power rating and thermal test pending

Assumptions and critique: guide walls are treated as uniform copper, the interior as smooth, and the guide as air-filled. Real surface roughness, plating, flange gaps, bends and transitions increase loss and can worsen matching. Permittivity is effectively that of air, but humidity/ contamination and any dielectric support can perturb fields. Analytical formulas predict cutoff accurately for ideal dimensions; field simulation or measurement is required for discontinuities and precise attenuation.
Tolerance: fc10 is inversely proportional to a. A +5% broad-wall error reduces cutoff by about 4.8% to roughly 6.25 GHz; a −5% error increases it to about 6.90 GHz. Both remain below the 9.5 GHz operating band. The TE20 cutoff is also inversely proportional to a and may shift by roughly ±5%; even a 5% lower TE20 cutoff is about 12.45 GHz, still above 10.5 GHz. However, flange gaps can materially affect S11 and must be checked experimentally.

EMC and safety plan: bond the enclosure and flanges, inspect conductive contact surfaces, use correct waveguide gaskets and connector transitions, and route RF away from sensitive electronics. Measure shielding leakage with a suitable probe/receiver, check grounding continuity, and perform an RF exposure/power-density assessment at accessible locations under maximum operating conditions. Confirm peak power and breakdown limits using component ratings.
Improvement plan: if measured insertion loss exceeds 0.20 dB or return loss falls below 20 dB, rework the transition taper, improve flange alignment, and tune a matching feature. Re-test the full band with a VNA and compare before/after S-parameter sweeps.

Final Engineering Decision (10 marks)

Initial design: NO. The preliminary total insertion-loss estimate is 0.242 dB, exceeding the 0.20 dB requirement, and the transition loss dominates.
Final improved design: YES, conditionally. The estimated total insertion loss is 0.104 dB, the TE10 mode covers 9.5–10.5 GHz with a 2.61 GHz margin to the next-mode cutoff at the upper band edge, and the matching target is return loss ≥25 dB (VSWR ≤1.12). Adoption is conditional on verifying the assumed transition loss, return loss, shielding and power rating by simulation and measurement. If tests fail, optimize the transition and flange interface before deployment.

Conclusion

WR-90 rectangular waveguide is a strong candidate for a 10 GHz civil weather-radar feed because it provides a defined TE10 propagation mode, low expected conductor loss, good power handling and shielding. The analytical design supports single-mode operation over the chosen band. A transition-focused redesign is predicted to satisfy the insertion-loss requirement, but measured S-parameters and EMC/power tests are necessary for final sign-off.

References

David M. Pozar, Microwave Engineering, 4th ed., Wiley.

R. E. Collin, Foundations for Microwave Engineering, 2nd ed., IEEE Press.

Constantine A. Balanis, Advanced Engineering Electromagnetics, 2nd ed., Wiley.

IEC 60153 series, Hollow metallic waveguides (applicable part to be selected for the purchased guide).

Manufacturer datasheet for WR-90 (WG16) waveguide, flanges and transitions; consult the actual selected supplier for certified band and power ratings.

Assignment source: ECA13 – Microwave Engineering, CO1 Application-Based Assessment, “Analysis, Design and Verification of a Microwave Transmission Line and Network for a Selected Application.”